EX NO: 27

CALCULATION OF PROCESSING GAIN AND GENERATION OF PN SEQUENCES FOR SPREAD SPECTRUM COMMUNICATION USING MATLAB SIMULATION

Objectives

1. To generate a pseudo-noise (PN) sequence using MATLAB for spread spectrum communication.
2. To determine the processing gain from the spreading factor or chip-to-bit ratio.
3. To perform spreading of a binary data sequence using the generated PN sequence.
4. To observe and analyze the characteristics of the generated PN sequence and spread signal.
5. To verify the relationship between processing gain, spreading factor, and bandwidth expansion.
6. To develop MATLAB-based skills for analyzing basic spread spectrum parameters.

CODE

```
clc;
clear;
close all;

% Processing Gain
Rb = 1000;       % Data rate
Rc = 10000;      % Chip rate

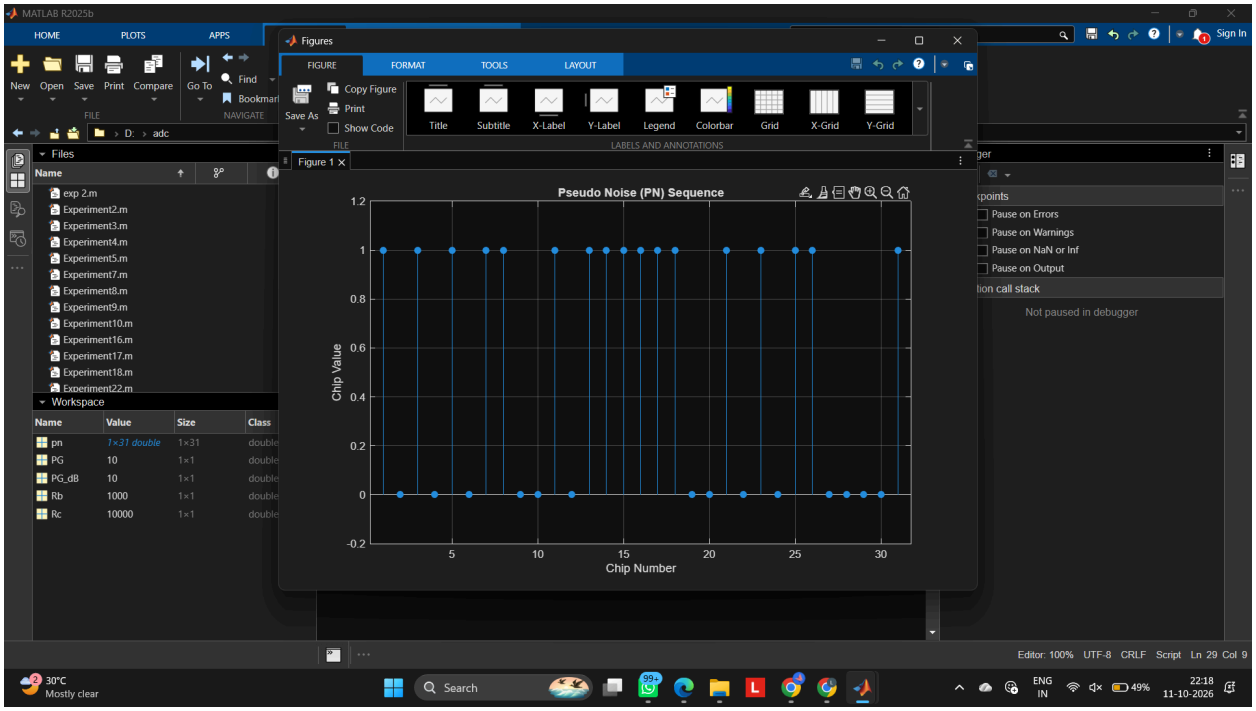
PG = Rc/Rb;
PG_dB = 10*log10(PG);

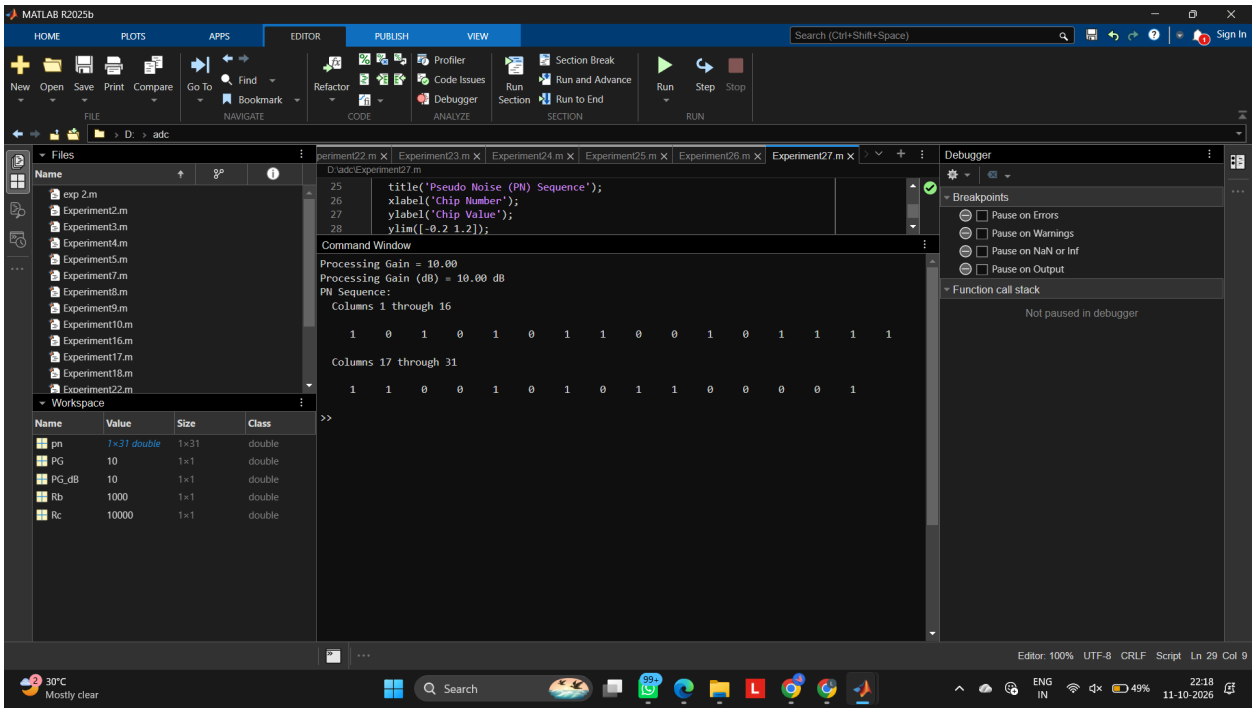
fprintf('Processing Gain = %.2f\n', PG);
fprintf('Processing Gain (dB) = %.2f dB\n', PG_dB);

% Generate PN Sequence
pn = randi([0 1], 1, 31);

% Display PN sequence
disp('PN Sequence:');
disp(pn);
% Plot PN sequence
figure;
stem(1:31, pn, 'filled');
title('Pseudo Noise (PN) Sequence');
xlabel('Chip Number');
ylabel('Chip Value');
ylim([-0.2 1.2]);
grid on;
```

OUTPUT


Pseudo Noise (PN) Sequence
Chip Value
Chip Number
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
5
10
15
20
25
30


Processing Gain = 10.00
Processing Gain (dB) = 10.00 dB
PN Sequence:
Columns 1 through 16
1 0 1 0 1 0 1 1 0 0 1 0 1 1 1 1
Columns 17 through 31
1 1 0 0 1 0 1 0 1 1 0 0 0 0 1